

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 1: Choice, Preference, and Utility

Notes on the Chapter

In my experience, this is a very difficult first chapter for students who have never taken a “serious” (theorem–proof) mathematics course, because it takes a while to learn how to work in fairly abstract categories. I have toyed with the idea of starting with a chapter on consumer demand, to keep things concrete, and then come back to these issues, but I’ve never found this approach to be satisfactory. Instead, I tell students who struggle to review (if that is the right word) the chapter on indifference curves and consumer demand from their intermediate micro textbook, to fix one concrete image. Depending on your audience, you might want to begin your course a bit more gently by giving an introductory/review lecture on intermediate micro style demand theory, then point out how the first few chapters develop these ideas more formally.

Notes on the Problems

The solutions to problems not listed here are given in the *Student's Guide*. Asterisks in front of problem numbers mean that their proofs are found in the *Guide*.

The levels of difficulty of the problems in this chapter will be apparent in most cases. Problem 1.4 is deceptively hard; please read the solution before you assign it. Problem 1.16 is hard in parts; the last part is very hard. For both problems, you might wish to tell students what the answers are and challenge them to find proofs/counterexamples. Finally, Problem 1.15 is very important preparation for Chapter 4, and students should at least be told to try it and then carefully read the solution given in the *Guide*.

Solutions to Non-Starred Problems

- 1.2. It is clear that c^* has the nonemptiness property, since for all (finite) nonempty A , $c_{\text{Larry}} \neq \emptyset$, and $c_{\text{Larry}}(A) \subseteq c^*(A)$.

But c^* does not necessarily satisfy choice coherence. Suppose $X = \{x, y, z\}$, and Larry's choice function is specified by $c_{\text{Larry}}(\{x, y, z\}) = \{x\}$ and $c_{\text{Larry}}(\{y, z\}) = \{y\}$, while $c_{\text{Moe}}(\{x, y, z\}) = \{z\}$ and $c_{\text{Moe}}(\{x, y\}) = \{x\}$. (Why are these enough to specify their choice functions? Because these choices, together with choice coherence, specifies choice in all other instances. It is easiest to see this by noting that the first choice by Larry implies $x \succ_{\text{Larry}} y$ and $x \succ_{\text{Larry}} z$, and the second implies $y \succ_{\text{Larry}} z$, which gives all of Larry's choices, for the induced preferences for Larry.) Then $c^*(\{x, y, z\}) = \{x, z\}$, but $c^*(\{y, z\}) = \{y, z\}$ (since $c_{\text{Moe}}(\{y, z\}) = \{z\}$). This is a clear violation of choice coherence.

■ 1.4. In one direction, we can make a tight connection: Given c^* , $x \succeq^* y$ if and only if $x \in c^*(\{x, y\})$. To prove this, suppose first that $x \succeq^* y$, so that either $x \succeq_{\text{Larry}} y$ or $x \succeq_{\text{Moe}} y$. In the first case $x \in c_{\text{Larry}}(\{x, y\})$, thus $x \in c^*(\{x, y\})$, and similarly in the second case. Conversely, if $x \in c^*(\{x, y\})$, then either $x \in c_{\text{Larry}}(\{x, y\})$ or $x \in c_{\text{Moe}}(\{x, y\})$. In the first case, $x \succeq_{\text{Larry}} y$, and thus $x \succeq^* y$, and similarly in the second case.

However knowing $\succeq^*$ is insufficient to construct c^* for more-than-two-element sets. Specifically, consider two cases:

Case 1: $x \sim_{\text{Larry}} y \sim_{\text{Larry}} z$. Then regardless of what Moe thinks, $\succeq^*$ has every ranking: $w \succeq^* w'$ for $w, w' = x, y, z$. Also, regardless of the choices made by Moe, $c^*(A) = A$.

Case 2: $x \succ_{\text{Larry}} y \succ_{\text{Larry}} z$ and $z \succ_{\text{Moe}} y \succ_{\text{Moe}} x$. It will take a moment, but it isn't hard to see that this also gives $\succeq^*$ that has every ranking. That is, this gives the same $\succeq^*$ as Case 1. But in this case $c^*(\{x, y, z\}) = \{x, z\}$.

We can prove that $x \in c^*(A)$ implies $x \succeq^* z$ for all $z \in A$ —the proof is very simple—but the converse fails, as the example shows.

■ 1.5. Suppose c satisfies choice coherence and finite nonemptiness. Suppose $x \in c(A)$ and $x \in B \subseteq A$. Since B is finite, some $y \in c(B)$, fix one such y . Since $y \in B \subseteq A$, $y \in A$. Apply the contrapositive form of choice coherence to conclude that $x \in c(B)$. That gives (α) .

Suppose c satisfies choice coherence (and finite nonemptiness). Suppose $y \in B$ and, for $B \subseteq A$, $y \in c(A)$. Suppose x is any member of $c(B)$. The contrapositive form of choice coherence immediately tells us that $x \in c(A)$. That is (β) .

Suppose c satisfies Sen's (α) and (β) . Suppose $x, y \in A \cap B$, $x \in c(A)$, and $y \in c(B)$. Since $A \cap B \subseteq A$ and $x \in A \cap B$, Sen's (α) implies that $x \in c(A \cap B)$. But $y \in A \cap B \subseteq B$, and $y \in c(B)$, so Sen's (β) implies that $c(A \cap B) \subseteq c(B)$, which means that $x \in c(B)$. Symmetrically, $y \in c(A)$, which is the contrapositive form of choice coherence.

Note that we needed finite nonemptiness (and the assumption that the sets involved are finite) only in showing that choice coherence implies (α) .

■ 1.8. Suppose $\succeq$ is complete and transitive. The set $\text{NBT}(x)$ is defined as $\text{NBT}(x) = \{y \in X : x \succeq y\}$. But $x \succeq x$ for all x , because $\succeq$ is complete. Therefore, $x \in \text{NBT}(x)$.

If $x \succeq y$ and $z \in \text{NBT}(y)$, then $y \succeq z$, hence $x \succeq y \succeq z$ which implies $x \succeq z$ by transitivity of $\succeq$. Therefore, $z \in \text{NBT}(x)$. This shows that $\text{NBT}(y) \subseteq \text{NBT}(x)$. Conversely, if $\text{NBT}(y) \subseteq \text{NBT}(x)$, we know from the first paragraph that $y \in \text{NBT}(y)$, hence $y \in \text{NBT}(x)$, hence $x \succeq y$.

If $x \succeq y$ and $y \not\succeq x$, then $\text{NBT}(y) \subseteq \text{NBT}(x)$ by the previous paragraph. We also know that $x \in \text{NBT}(x)$ by the first paragraph, but since $y \not\succeq x$, $x \notin \text{NBT}(y)$. Therefore, $\text{NBT}(y)$ is a proper subset of $\text{NBT}(x)$.

Take any two x and y . By completeness, either $x \succeq y$ or $y \succeq x$, and by the second paragraph, this implies that either $\text{NBT}(y) \subseteq \text{NBT}(x)$ or $\text{NBT}(x) \subseteq \text{NBT}(y)$; the no-better-than sets nest.

■ 1.9. Suppose $\succ$ is asymmetric and negatively transitive. Take any two x and y . By asymmetry, either not $x \succ y$, in which case $y \succeq x$, or not $y \succ x$, in which case $x \succeq y$. Hence $\succeq$ is complete. Suppose $x \succeq y \succeq z$. This means that not $y \succ x$ and not $z \succ y$. Now negative transitivity says that if $z \succ x$, then either $y \succ x$ or $z \succ y$; since both of these are false, it cannot be that $z \succ x$. (In other words, “not $y \succ x$ and not $z \succ y$ implies not $z \succ x$ ” is the *contrapositive* of negative transitivity: The statements “A implies B” and “not B implies A” are equivalent statements, called the contrapositives of one another.) Thus $x \succeq z$, and $\succeq$ is transitive.

Suppose $\succ$ is asymmetric and negatively transitive, and define $\succeq$ by $x \succeq y$ if not $y \succ x$. Define $\succ'$ by $y \succ' x$ if $y \succeq x$ and not $x \succeq y$. That is, $y \succ' x$ if not $x \succ y$ and not not $y \succ x$, which is not $x \succ y$ and $y \succ x$. But if $y \succ x$, not $x \succ y$ follows from asymmetry, so this says that $y \succ' x$ if $y \succ x$. On the other hand, if $y \succ x$, then not $x \succ y$, hence both not not $y \succ x$ and not $x \succ y$, hence not $x \succeq y$ and $y \succeq x$, hence $y \succ' x$. $\succ$ is the same as $\succ'$.

And suppose $x \sim y$. This is true if not $x \succ y$ and not $y \succ x$, which implies $y \succeq x$ and $x \succeq y$, which implies $x \sim' y$. Conversely, $x \sim' y$ if $x \succeq y$ and $y \succeq x$, which is not $y \succ x$ and not $y \succ x$, which is $x \sim y$. Hence $\sim$ and $\sim'$ are the same.

■ 1.12. To take the easy part first, the function $v(x) = x$ is a continuous function that represents $\succeq$. As for the function produced in the proof of Proposition 1.12, this is a strictly increasing, right-continuous function that jumps discontinuously at every rational number. That is, $u(0) = \frac{1}{2}$, since $\text{NBT}(0) \cap X^*$ is $\{0\}$. $u(1) = 1$, since $\text{NBT}(1) \cap X^*$ is all the rationals. But for any $x < 1$, $u(x) < 0.75$, since $\text{NBT}(x)$ necessarily lacks the element 1 (and other rationals, besides), and in the sum that defines u , the element 1 gets the “weight” 0.25. If you are skilled in real variables, you should be able to prove that u is indeed continuous from the right and continuous at every irrational number, but if z is rational and $d(z)$ is the weight associated with z , then $\lim_{x \rightarrow z} u(x) = u(z^-) = u(z) - d(z)$.

■ 1.14. (a) Under the conditions given, for all x and y from X , $x \succeq_u y$ if and only if $u(x) \geq u(y)$ if and only if $f(u(x)) \geq f(u(y))$ (since f is strictly increasing on $u(X)$) if

and only if $v(x) \geq v(y)$ if and only if $x \succeq_v y$.

(b) We argue by contrapositive. Suppose f is not strictly increasing on $u(X)$. That is, there exist x and y from X such that $u(x) > u(y)$ but $f(u(y)) \geq f(u(x))$. But then $u(x) > u(y)$ implies $x \succ_u y$ implies $y \not\succeq_u x$, while $v(y) = f(u(y)) \geq f(u(x)) = u(x)$ implies $y \succeq_v x$, so $\succeq_u$ and $\succeq_v$ are not identical.

(c) Since $v(x) = f(u(x))$, taking x to be 1 shows that $f(1) = 1$. And taking $x = 1 + 1/n$ (for positive integer n), we have that $u(x) = 2 + 1/n$ and $v = 1 + 1/n$, so we know that $f(2 + 1/n) = 1 + 1/n$. Now suppose f is strictly increasing. This implies that $f(2) < f(2 + 1/n) = 1 + 1/n$, and since n is arbitrary here, this implies that $f(2) \leq 1$. But this cannot be if f is strictly increasing, since $f(1) = 1$.

(d) Fix $x^0 \in X$, and let $r = u(x^0)$. Then $X_r = \{x \in X : u(x) \leq r\} = \{x \in X : u(x) \leq u(x^0)\} = \{x \in X : x^0 \succeq_u x\} = \{x \in X : x^0 \succeq_v x\}$ (since $\succeq_u \equiv \succeq_v$) $= \{x \in X : v(x) \leq v(x^0)\}$. For if $f(r) = \sup\{v(x) : x \in X_r\}$, it is apparent that $f(r) = v(x^0)$; for all x in the set over which the supremum is taken, $v(x) \leq v(x^0)$, and x^0 is in the set. This implies that $v = f \circ u$ on $u(X)$.

Part b now applies; we know that f is strictly increasing on $u(X)$.

To show that it is nondecreasing on all of R , take any two real numbers r and r' with $r \geq r'$. It is evident from the definition that $X_r \supseteq X_{r'}$, and therefore when we take the suprema that define $f(r)$ and $f(r')$, the first is at least as large, as we are taking a sup over a (weakly) larger set.

But there are two potential problems: X_r might be empty, in which case the supremum is (by convention) $-\infty$. And if X_r contains a sequence $\{x_n\}$ such that $\lim_n v(x_n) = \infty$, then the supremum is ∞ .

These are not idle possibilities. Suppose $X = R$, and let $v(x) = x$; that is, we are looking at $\succeq_v$ (and $\succeq_u$) being the "natural order" on the real line. And let $u(x) = \arctan(x)$ (where we are taking the principal root of the arc tangent; that is $u(x)$ lies (strictly) between $-\pi/2$ and $\pi/2$). Of course, the arc tangent function is strictly increasing, so $\succeq_u$ is indeed identical with $\succeq_v$. And then, for $r = -\pi$, X_r is empty, and we get $f(-\pi) = -\infty$, while for $r = \pi$, $X_r = R$, and when we take the supremum, we get $f(\pi) = \infty$. The problem is obvious: v "takes up" all the real numbers, so for f to be nondecreasing, we need $-\infty$ to be a value of f for r below $\inf u(x)$ and we need ∞ for r above $\sup u(x)$. To tie up all the loose ends, you might want to prove that if g is any extended real-valued function such that $v \equiv g \circ u$ (for these specific u and v), if g is nondecreasing, then $g(r) = -\infty$ for all $r \leq -\pi/2$ and $g(r) = \infty$ for $r \geq \pi/2$. Alternatively, prove that g cannot be both nondecreasing and real valued in this case.

■ 1.15. The proof of Proposition 1.5 never deals in subsets of X of size greater than three, except where it deals with a fixed subset that, in the context of this problem, is from $\mathcal{A}$ and so has nonempty choice. That proof works without modification.

■ 1.17. (a) Suppose to the contrary that $x \succ y$ and either $x \sim y$ or $y \sim x$. Since $\sim$ is symmetric, $x \sim y$ implies $y \sim x$, so we can assume $x \succ y$ and $y \sim x$. By property (3), these imply $x \succ x$, contradicting the supposed asymmetry of $\succ$.

(b) Weak preference defined in this fashion is certainly transitive: If we have x, y , and z such that $x \succeq y$ and $y \succeq z$, then there are four possibilities: (1) $x \succ y$ and $y \succ z$, in which case the transitivity of $\succ$ implies $x \succ z$ and thus $x \succeq z$. (2) $x \sim y$ and $y \succ z$, in which case the transitivity of $\sim$ implies $x \sim z$ and thus $x \succeq z$. (3) $x \sim y$ and $y \sim z$, in which case property (4) implies $x \succ z$ and thus $x \succeq z$. (4) $x \succ y$ and $y \sim z$, in which case property (3) implies $x \succ z$ and thus $x \succeq z$.

On the other hand, weak preference so defined is not necessarily complete. The easiest way to show this is to enlist part (d), so we put off an example until we have finished part (d).

Although you weren't asked to show this, it is worthwhile to note that because $\sim$ is reflexive, so is $\succeq$.

(c) Suppose $\succeq$ is transitive and reflexive and we define $\succ$ and $\sim$ by

$$x \succ y \quad \text{if} \quad x \succeq y \text{ and not } y \succeq x, \quad \text{and} \quad x \sim y \quad \text{if} \quad x \succeq y \text{ and } y \succeq x.$$

I assert that these two binary relations then have properties (1) through (4). Taking them in order:

(1) $x \succ y$ if $x \succeq y$ and not $y \succeq x$, which implies that not $y \succ x$: $\succ$ is asymmetric by definition. And if $x \succ y$ and $y \succ z$, then $x \succeq y$, $y \succeq z$, not $y \succeq x$, and not $z \succeq y$. Transitivity of $\succeq$ implies $x \succeq z$. Suppose that $z \succeq x$. Then by transitivity of $\succeq$, $z \succeq y$, a contradiction. Therefore, not $z \succeq x$, and $x \succ z$.

(2) According to the definition of $\sim$, $x \sim y$ if $x \succeq y$ and $y \succeq x$. Reflexivity of $\succeq$ implies reflexivity of $\sim$. Symmetry of $\sim$ follows from symmetry of its definition. Transitivity follows by transitivity of $\succeq$: If $x \sim y$ and $y \sim z$, then $x \succeq y \succeq z \succeq y \succeq x$, thus $x \succeq z \succeq x$.

(3) If $x \succ y$ and $y \sim z$, then $x \succeq y \succeq z$, thus $x \succeq z$ by transitivity of $\succeq$. Suppose that $z \succeq x$. Then $y \succeq z \succeq x$, implying $y \succeq x$ by transitivity, which would contradict $x \succ y$. Thus not $z \succeq x$, and hence $x \succ z$.

(4) If $x \sim y$ and $y \succ z$, then $x \succeq y \succeq z$ and $x \succeq z$ by symmetry. Suppose $z \succeq x$. Then $z \succeq x \succeq y$, implying $z \succeq y$ by transitivity, which would contradict $y \succ z$. Thus not $z \succeq x$, and hence $x \succ z$.

(d) If $\succ$ and $\sim$ are defined from U in this fashion, then they satisfy all four properties:

(1) $x \succ y$ if $U(x) > U(y) + 1$, which implies $U(y) \not> U(x) + 1$, thus not $y \succ x$. Thus $\succ$ is asymmetric. And $x \succ y \succ z$ if $U(x) > U(y) + 1$ and $U(y) > U(z) + 1$, which together

imply $U(x) > U(z) + 2 > U(z) + 1$, thus $x \succ z$. Thus $\succ$ is transitive.

(2) Reflexivity, symmetry, and transitivity of $\sim$ all follow immediately from the same properties of $=$ for real numbers.

(3) If $x \succ y$ and $y \sim z$, then $U(x) > U(y) + 1$ and $U(y) = U(z)$, thus $U(x) > U(z) + 1$, and $x \succ z$.

Property (4) is proved almost precisely as is property (3).

Now we can finish part (b): Suppose $X = \{0, .1\}$, let $U(x) = x$, and define $\succ$ and $\sim$ from U as in part (d). We know from part (d) that $\succ$ and $\sim$ satisfy the four properties. And by their definitions from U , none of $0 \succ .1$, $.1 \succ 0$, $0 \sim .1$, and $.1 \sim 0$ hold. Thus neither $0 \succeq .1$ nor $.1 \succeq 0$. In this example, $\succeq$ is not complete.

(Aside to the curious student: Why did we wait until finishing with part (d) to give this simple counterexample to completeness? The reason is that it is easy to give the example $X = \{0, .1\}$, $0 \sim 0$, $.1 \sim .1$, and no other relation holds, which (clearly) gives an incomplete $\succeq$. But then we must verify that for this X , $\sim$, and $\succ$, the four properties hold. This isn't hard, but it is tedious, and since we had to show (d) anyway, there was no need to do the extra work.)

Just to summarize where we are: Properties (1) through (4) for $\succ$ and $\sim$ are equivalent to having symmetric and transitive (but not necessarily complete) weak preferences. (Have we really shown this? Not quite. To lock up the equivalence of the two approaches, we need to show that if we start with $\succ$ and $\sim$ with the four properties and define $\succeq$ as in part (b), and then redefine $\succ$ and $\sim$ in the fashion of this problem, we are back to where we began, and vice versa. Happily, these things are true, and if you want to be a fanatic about this alternative to the standard theory, you are cordially invited to give the details.) These properties are implied by the representation in part (d). Now we want to know whether the properties imply the representation.

(e) They do not. Suppose that $X = \{0, 1, \dots, 10\} \times \{\text{red}, \text{blue}\}$, so that a typical $x \in X$ is of the form (z, c) , where z is an integer between zero and ten, and c is either the color red or the color blue. In other words, think of X as consisting of two sets of integers, one in red ink and the other in blue. Define $V((z, c)) = z$, and define $(z, c) \succ (z', c')$ if $z > z' + 1$ and $c = c'$, and $(z, c) \sim (z', c')$ if $z = z'$ and $c = c'$. In other words, two numbers of different colors are incomparable, and numbers of the same color compare as in the basic representation.

I leave it to you to verify that the four properties hold. This is relatively easy, once we have part (d), since throughout we only get comparisons of either sort of same-colored numbers. But there is no single function U that gives the representation. To see this, suppose that there did exist such a single function. Because $(3, \text{red}) \succ (1, \text{red})$, we would need to know that $U((3, \text{red})) > U((1, \text{red})) + 1$. Similarly, $U((4, \text{blue})) > U((2, \text{blue})) + 1$. Now by properties of real numbers, either $U((3, \text{red})) \geq U((4, \text{blue}))$ or $U((4, \text{blue})) > U((3, \text{red}))$. In the former case, $U((3, \text{red})) > U((2, \text{blue})) + 1$, which would

imply that $(3, \text{red}) \succ (2, \text{blue})$, which isn't true. And in the latter case, $U((4, \text{blue})) > U((1, \text{red})) + 1$ which would imply $(4, \text{blue}) \succ (1, \text{red})$, which also isn't true. Hence no single function U can work.

(f) In addition to the four properties, we must add the following two variations on negative transitivity:

(5) If $x \succ y$ and $z \succ w$, then either $x \succ w$ or $z \succ y$ (or both).

(6) If $x \succ y \succ z$, then for all w , either $x \succ w$ or $w \succ z$.

Property (5) can be paraphrased: If there is a large enough gap between x and y (large enough to give strict preference), and a large gap between z and w , then either z is enough better than y or x is enough better than z to see the requisite gap. Property (6) is paraphrased, if the gap between x and z is "double-sized," in the sense that y fits between them with strict preference each way, then negative transitivity in the usual sense holds for x and z .

If we have the representation described in part (d), properties (5) and (6) must hold. For (5), mimic the counterexample constructed in part (e). Suppose we have a representing utility function U . If $x \succ y$ and $z \succ w$, then $U(x) > U(y) + 1$ and $U(z) > U(w) + 1$. Either $U(x) \geq U(z)$, in which case $U(x) > U(w) + 1$ and thus $x \succ w$, or $U(z) \geq U(x)$, in which case $U(z) > U(y) + 1$, and thus $z \succ y$. For (6), if $x \succ y \succ z$, then $U(x) > U(z) + 2$, and a comparison of $U(w)$ with $U(z) + 1$ gives: If $U(w) > U(z) + 1$, then $w \succ z$, while if $U(w) \leq U(z) + 1$, then $U(x) + 1 > U(w)$ and $x \succ w$.

Moreover, the six properties imply the existence of a representing function U (for finite X !!!!).¹ I will sketch a proof which is a bit long-winded but which most clearly reveals what is going on in this theory.

Fix a finite set X on which are defined $\succ$ and $\sim$ satisfying the six properties. For each $x \in X$, define

$$W(x) = \{z \in X : x \succ z\} \quad \text{and} \quad B(x) = \{z \in X : z \succ x\}.$$

I assert that the W and B sets nest. That is, for all x and y , either $B(x) \subseteq B(y)$ or $B(y) \subseteq B(x)$, and either $W(x) \subseteq W(y)$ or $W(y) \subseteq W(x)$. These properties follow from (5): Suppose neither $B(x) \subseteq B(y)$ nor $B(y) \subseteq B(x)$. Then there exists $x' \in B(x) \setminus B(y)$ and $y' \in B(y) \setminus B(x)$. But then $x' \succ x$, $x' \not\succ y$, $y' \succ y$, and $y' \not\succ x$, a contradiction to (5). (A similar argument shows that the W sets nest.)

Moreover, the two sets of nests are connected as follows:

$$\text{If } W(x) \subseteq W(y) \text{ and } W(x) \neq W(y), \text{ then } B(y) \subseteq B(x). \quad (\star)$$

¹ I am certain that this result is somewhere in the literature, but I don't have a citation. I would be grateful if someone could provide me with one.

To see this, suppose to the contrary that $W(x) \subseteq W(y)$, $W(x) \neq W(y)$, and $B(y) \not\subseteq B(x)$. The first two imply the existence of $z \in W(y) \setminus W(x)$ and the last some $w \in B(y) \setminus B(x)$. That is, $w \succ y \succ z$, but neither $w \succ x$ nor $x \succ z$, a contradiction to (6).

For each $x \in X$, let $b(x)$ be the number of elements in $B(x)$ and let $w(x)$ be the number of elements in $W(x)$. Define two binary relations, $\succ$ and $\sim$, by

$$\begin{aligned} x \succ x' & \quad \text{if} \quad w(x) > w(x') \quad \text{or} \quad w(x) = w(x') \text{ and } b(x) < b(x'), \\ \text{and} \quad x \sim x' & \quad \text{if} \quad w(x) = w(x') \text{ and } b(x) = b(x'). \end{aligned}$$

Because of the nesting properties of the W and B sets, it is easy to see that

$$x \sim x' \quad \text{if and only} \quad W(x) = W(x') \text{ and } B(x) = B(x'),$$

$$\text{and} \quad x \succ x' \quad \text{if and only} \quad W(x) \supset W(x') \quad \text{or} \quad W(x) = W(x') \text{ and } B(x) \subset B(x'),$$

where for the rest of this solution we use $\subset$ (respectively, $\supset$) to mean “subset (resp., superset) of, but not equal to.” (The details of this step are left to you.)

Also left to you is a demonstration of the fact that $\succ$ and $\sim$ have properties (1) through (4) and, in addition, for every pair x and y , one of $x \succ y$ or $y \succ x$ or $x \sim y$ must hold. (For some of the details, see the solution to Problem 1.14.)

Therefore (and there are a lot of details in this *therefore*), we can enumerate X as $\{x_1, x_2, \dots, x_N\}$, where N is the cardinality of X , such that for each $n = 1, \dots, N-1$, either $x_n \succ x_{n+1}$ or $x_n \sim x_{n+1}$.

I assert that there exists a function $V : X \rightarrow R$ such that (a) $V(x) = V(y)$ if and only if $x \sim y$, (b) $V(x) > V(y)$ if and only if $x \succ y$, (c) $V(x) > V(y) + 1$ if $x \in B(y)$ (equivalently, if $y \in W(x)$), and (d) $V(y) < V(x) < V(y) + 1$ if $x \succ y$ but $x \notin B(y)$.

To construct such a function, proceed as follows: Enumerate the set X as $\{x_1, x_2, \dots, x_N\}$ as above. Let $V(x_1) = 0$. Then take each x_n in turn, defining V for it so that $V(x_n) = V(x_{n+1})$ if $x_n \sim x_{n+1}$ and $V(x_n) > V(x_{n+1})$ if $x_n \succ x_{n+1}$. As long as we do this, we have (a) and (b). To show that (c) and (d) can be done as well, suppose we have defined V as far as x_n so that (a) to (d) hold for $\{x_1, \dots, x_n\}$. Take x_{n+1} . Either $x_n \sim x_{n+1}$, or $x_n \succ x_{n+1}$.

If $x_n \sim x_{n+1}$, we let $V(x_{n+1}) = V(x_n)$. Then properties (c) and (d) hold for the augmented set because $x_n \sim x_{n+1}$ implies $B(x_n) = B(x_{n+1})$, hence these properties holding for x_n ensure they hold as well for x_{n+1} .

If $x_n \succ x_{n+1}$, identify that element x_m in the chain x_1 to x_n such that $x_m \in B(x_{n+1})$ but $x_{m+1} \notin B(x_{n+1})$. (The special cases where $m = n$ and where $x_1 \notin B(x_{n+1})$ are left to you to handle.) Then $x_{n+1} \in W(x_m)$, and by the nesting property of the W sets, $x_{n+1} \in W(x_k)$ for all $k \leq m$, while $x_{n+1} \notin W(x_k)$ for all $k \geq m+1$. Since $W(x_m) \supset W(x_{m+1})$

(by virtue of x_{n+1}), it follows that $x_m \succ x_{m+1}$, and thus $V(x_m) > V(x_{m+1})$. Moreover, we assert that $x_{m+1} \notin B(x_n)$. (This is the key step in the proof, so here is where you must pay attention!) To see this, note that $x_n \succ x_{n+1}$ means that either $W(x_n) = W(x_{n+1})$ and $B(x_n) \subset B(x_{n+1})$, in which case we get the desired conclusion, or $W(x_n) \supset W(x_{n+1})$, in which case the desired conclusion follows from an application of property $(\star)$. Thus $V(x_{m+1}) - V(x_n) < 1$. If $x_n \in W(x_m)$, then $V(x_m) - V(x_n) > 1$. In this case, let $V(x_{n+1}) = V(x_n) - [1 + V(x_n) - V(x_{m+1})]/2$. This is a bit less than $V(x_n)$, but not so much less that we violate $V(x_{m+1}) - V(x_{n+1}) < 1$. And if $x_n \notin W(x_m)$, then $V(x_n) > V(x_m) - 1$. If we set $V(x_{n+1}) = [V(x_m) + V(x_{m+1})]/2 - 1$, we are assured that $V(x_{n+1}) < V(x_n)$, $V(x_{n+1}) < V(x_m) - 1$, and $V(x_{n+1}) > V(x_{m+1}) - 1$. We have successfully extended V . Proceeding in this fashion (since X is finite), we can produce the desired function V .

The function V is almost, but not quite, the desired function U . The problem is that $V(x) = V(y)$ if $x \sim y$, and we want $U(x) = U(y)$ only if (as well) $x \sim y$. It is clear that $x \sim y$ implies $x \sim y$, but the reverse is false. But to get U from V is simple. Let

$$\delta = \min \{ \min\{V(x) - V(y) - 1 : x \succ y\}, \min\{1 - V(x) + V(y) : x \succ y, x \not\sim y\} \}.$$

By construction $\delta > 0$, and if we create a function U by adding or subtracting no more than $\delta/3$ to each value of V , we preserve the relationship that $x \succ y$ if and only if $U(x) > U(y) + 1$. Thus we can “adjust” V to get U so that $U(x) = U(y)$ if and only if $x \sim y$ (using property (2) to show that $\sim$ is an equivalence relationship) while preserving $x \succ y$ if and only if $U(x) > U(y) + 1$. That finishes the proof.